How colonization economies work

A plain-language guide to how Raven Colonial estimates market economy percentages for your ports. This describes the same rules as the technical economy model (companion: economy-model.md), without implementation detail.

What you see in the app: each dockable port gets a percentage per economy type (Agriculture, Refinery, Military, and so on). Those numbers are built from the port's **home body**, **system resources**, and **links** to other facilities and ports across the system. The economy table's audit trail is a **ledger**: every line is a +5%, +40%, +80%, or +100% step; the displayed percent is the sum (225% means 2.25 in the data).

External check: where stations are complete in-game, the architect UI can compare estimates against **Spansh** market snapshots.

Two docs: this guide explains *what happens* in player terms. economy-model.md (companion: economy-model.md) maps the same rules to source files and formulas.

The big picture

Think of each port's economy as a **ledger** of +5%, +40%, +80%, and +100% steps. The final percentage is the sum of every line that applies to that port.

Three mechanisms feed the ledger:

What the port earns on its own

- Body type (e.g. Rocky → Refinery)
- Body features (e.g. Organics → Agriculture)
- System resource level (e.g. Pristine)

Strong links — big contributions

- Nearby hubs and settlements
- Partner port on the same moon

Weak links — small +5% steps

- Facilities and ports elsewhere in the system

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Docked market %

Mechanism	Typical size	Modified by body conditions?
Own row (body + system buffs)	+40% or +100% chunks	Yes for agriculture boosts; yes for pristine/depleted on industry
Strong link	+40%, +80%, or +120% per source	Yes — especially agriculture and pristine boosts on each link
Weak link	+5% per source	**Never** — always flat five percent

The **link graph** in Market Links shows *candidates*. For agriculture, a **budget** may cap how many weak +5% steps actually apply even when many sources are listed.

Who gets calculated: by default only **completed** dockable sites. Planning mode can include planned builds — useful for previews, not for Spansh comparison.

What counts when you build

By default, only **completed** facilities affect the model. "Include incomplete" planning mode can add planned and under-construction sites up to a build-order limit — useful for preview, not for Spansh comparison.

Hubs and installations (relay, security post, space farm, refinery hub, and so on) have fixed roles: they feed **links** into ports; their own economy row comes from a facility table, not from the port pipeline below.

Step by step: how a colony port gets its numbers

1. **Body gifts** — The landable body adds whole economies from its type (rocky → refinery, earth-like → agriculture + tourism + …) and from signals (organics → agriculture, geological → industry, and so on).
2. **Local buffs** — If the port already has an economy from step 1, the system may add **+40%** for pristine/major resources (industry, refinery, extraction), organics (agriculture and hightech), or similar. Low or depleted resources can subtract 40% on those same industry types.
3. **Strong links** — Large chunks from supporting facilities on the same body and from partner ports (see below). Each strong contribution can receive **extra +40%** in a pristine system for refinery, industrial, and extraction — **once per strong link**, not once total.
4. **Weak links** — Every qualifying source elsewhere in the system adds **+5%** for its economy type (agriculture uses a budget cap).

5. **Finish** — The highest total becomes the port's **primary economy**; all non-zero lines are shown in the economy table.

Order matters for weak links: nearby same-body agriculture sources are counted first (agriculture only), then system-wide sources in **alphabetical order**. That is why a science hub's hightech can unlock a relay +5% on a starport when the relay's name sorts later.

Settlements (Odyssey farms and similar) use a simpler path: fixed agriculture percentage, local buffs, no strong or weak links applied *to* the settlement — but settlements can still *send* weak links to ports elsewhere when they are subordinate hubs in the link graph.

Specialized ports (refinery outpost, tourism starport, etc.) start at +50% surface or +100% orbital on their specialty, then follow the same link rules as general colonies.

Strong links in plain terms

Strong links are the main way a system specializes. A refinery hub, extraction settlement, or second port on a moon pushes a large slice of its economy into a dockable port.

Size by facility tier

Tier	Strong link size
Small (outpost, small hub, settlement)	+40%
Medium hub	+80%
Large hub / starport-tier source	+120%

Same moon, two ports — If you have a surface outpost and an orbital starport on one body, the surface port's body economies can **strong-link** into the orbital port. The orbital port does **not** duplicate agriculture buffs on its own row when a surface **colony port** is present; those conditions apply on the **link** instead.

Hub children — A refinery hub may list settlements underneath it. Those appear as **sub-strong** links into the body's main port (+40% / +80% from the child's tier), each able to pick up a pristine +40% boost in refinery/extraction/industrial.

Gas-giant farms — A space farm on a sibling moon **strong-links agriculture** into that moon's **main port only** (usually the orbital starport). A second outpost on the moon gets agriculture through **weak +5%** steps, not the farm's strong link.

Weak links in plain terms

Weak links are system-wide “background influence”: +5% per source, no bonuses or penalties on that five percent.

Who can send weak links

Sender	Behavior
Relay installation	Hightech +5% to ports across the system
Security installation	Military +5% to all ports not on its body
Unanchored space farm	Agriculture +5% to other bodies
Subordinate port or hub	Its primary economy type, outward
Body primary port	Agriculture outward; on the **star** body, only agriculture (not military/refinery weak export)
Military hub installation	Does not weak-link outward
Relay on a starport with no hightech	Link shows in UI only — no +5% on the market row (e.g. a large ornamental starport)
Relay on a starport that already has hightech	+5% applies on top (e.g. science hub weak link, then relay)

Weak links are processed in **alphabetical order** by source name. That order matters when a relay only applies after another source has already unlocked hightech on a starport.

Security posts add military +5% to ports on **other bodies** in the system. A security install on the same moon as your outpost does not weak-link military into that outpost — cross-body pool only.

Foreign stars: a surface colony with no local agriculture may only accept **one** agriculture weak link from each distant star’s subtree (farms on another star count once). Agriculture settlements are exempt from that cap.

Agriculture — the special case

Agriculture follows the same three-path pattern observed in multi-port and tidal-moon systems (e.g. subordinate surface **145%**, paired orbital primary **225%**).

On the port’s own market row

- **+100%** if the body has organics (and isn’t already an earth-like/water world).
- **+40%** organics buff on that row for most ports.

- **Exception:** an **orbital starport** paired with a **surface colony port** on the same body does **not** get those agriculture buffs on its own row — they appear on the **strong link** from the surface port instead.
- **No** icy or tidal **penalties** on the own row (subordinate surface ports keep +40% organics without −40% tidal on the docked row).

On strong links into this port

- Full **+40% / −40%** table: organics, earth-like/water world, icy body, tidal lock chain to star, and so on.
- Tidal penalty on a **link** cannot pull that link's contribution below **+10%**.

On weak links

- Flat **+5%** per farm or agriculture port, up to the port's agriculture **budget**.

Intuition: your outpost's **displayed** agriculture is "what grows here." **Penalties** for tide and ice apply when agriculture is **pushed through a link** into another port, not when you're listing the outpost's own row. Weak links are always a simple +5%.

Agriculture weak-link budgets (plain terms)

Not every farm in the link graph adds +5% to your port. The model caps total weak agriculture by a **budget** — think "how much distant farm influence this port can absorb." Tighter budgets apply when:

- A **strong** agriculture link already comes from the same moon (only a sliver of weak links left),
- The port is on a **metal-rich** or **icy** body without being an ag specialist,
- The port is an **orbital cluster** type with many subordinates (budget scales with count),
- The body is a **habitable ag-primary** world with a small colony strong link.

When the budget is exhausted, extra agriculture sources still appear in Market Links but audit shows **Skipped weak link … cap reached**. Typical default cap is **90%** from weak links alone (eighteen +5% steps) before other rules tighten it.

Refinery, industrial, and extraction — pristine systems

In **major** or **pristine** systems:

Where	+40% pristine bonus
Port's **own** refinery/industrial/extraction from the body	Once, on the intrinsic row
Each strong link that adds refinery/industrial/extraction	Again, per link

Example — orbital starport on a rocky moon in a pristine system:

- +100% rocky refinery (body)
- +40% pristine on that intrinsic
- +40% strong link from surface partner port → +40% pristine on **that** link
- +80% sub-strong from refinery hub on the moon → +40% pristine on **that** link
- **Total refinery 340%**

A port with **only one** refinery strong link gets one pristine boost on the link (e.g. 260% on a single-hub starport).

Hightech and tourism strong-link body bonuses (organics, earth-like, black hole, and so on) still apply **at most once** per port calculation, unlike pristine industry boosts.

Worked example — orbital starport refinery (340%)

Ledger line	Amount
Rocky body → refinery	+100%
Pristine system buff on that row	+40%
Strong link from same-body surface colony partner	+40%
Pristine buff on **that** strong link	+40%
Sub-strong from refinery hub child on the moon	+80%
Pristine buff on **that** sub-strong link	+40%
Total	**340%**

Each refinery **strong** contribution in a pristine system can pick up its own +40% — the body row is not the only beneficiary.

Multiple ports on one body

Frontier's docs discourage multiple ports per body; the model still covers it:

1. One **primary** port per body (orbital wins over surface when both exist).
2. **Subordinate** ports link to the primary and **share its weak-link pool**.
3. Surface colony → orbital colony **strong-links** body economies; orbital agriculture buffs may move to the link layer.
4. Space farms **strong-link** the primary only; subordinates rely on weak agriculture steps.

Building extra ports is fragile in-game; the model reflects that split between primary and converted/subordinate behavior.

Facilities you'll see often

Facility	Effect on ports
Relay	Weak hightech +5% system-wide (with starport rule above)
Security post	Weak military +5% system-wide
Space farm	Strong agriculture locally; weak agriculture to other bodies if not tied to a port on the same body
Refinery / industry hub	Strong link of that type into the body primary
Comms	Unlocks higher hightech on a science hub when complete on the body or parent chain
Military hub	Strong military locally; does not weak-link outward

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Documented rules vs tuning knobs

Documented rules come from community colonization references (Mega Guide, Update 3): body tables, strong/weak sizes, subordinate behavior, agriculture three-path model, relay and security scope.

Tuning knobs are extra caps fit to observed markets: agriculture weak-link **budgets** (how many +5% steps count), some settlement floors, and a few colony presets. When in doubt, the documented layer is authoritative; tuning exists where the public rules are silent.

Open gap: low/depleted resource **penalties** on industry may eventually move to strong-link-only (as agriculture icy/tidal already did). Today they still apply on the port's own industry row.

Reading the economy table

Each line in the audit trail is one ledger entry: "+40 Buff: body has BIO", "+80 Apply sub-strong link from: …", "+5 Apply weak link from: …".

Sum the lines for an economy type to get the displayed percentage (shown as 145% meaning 1.45 in the data).

Primary economy is whichever type has the highest total after all steps.

Representative scenarios

Scenario	What it exercises
Gas-giant cluster + ornamental starport	Relay UI-only when starport has no hightech; agriculture weak links from subordinate; security military cross-body; sibling-moon farm strong link
Multi-port tidal moon + star-body port	Surface/orbital colony pair; relay hightech after another weak source; pristine refinery stacking per strong link; security on star; anchored vs cluster farms

Use these patterns when validating link or buff rule changes.

Quick reference card

Question	Answer
How big is a weak link?	Always +5%
How big is a strong link?	+40% / +80% / +120% by source tier
Does tide hurt my outpost's ag % on its own row?	No −40% on own row; yes on strong links into other ports
Does pristine help refinery twice?	Yes — on body intrinsic and on **each** refinery strong link
Does relay always add hightech?	On outposts yes; on starports only if hightech already > 0 from another link
Who weak-links military from a security install?	Everyone not on that install's body
Can a star-body primary export refinery weak links?	No — star-body primary exports agriculture weak only
Do farms on another moon strong-link my outpost?	No — strong to that moon's **primary** port only

For modifiers tables, budget rule list, and module ownership, see economy-model.md (companion: economy-model.md).